

Jakub Halmeš

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Work Experience

AI Engineer | Chyron

Jul 2025 - present (Brno, Czech Republic), Full-time

Shifted to a more applied role focused on AI engineering and productionization. Designed and developed a gRPC image tagging service using VLMs (OpenAI API, HuggingFace), contributed to developing a pipeline for semantic retrieval (Qdrant).

Currently: working on evaluation of a multi-agent LLM system (Langchain, MLFlow). Exploring and streamlining the AI code review process as part of a company-wide AI Guild initiative.

Machine Learning Intern | Chyron

Jun 2022 - Jun 2025 (Brno, Czech Republic), Internship

A 3-month internship extended into a longer-term collaboration with approximately part-time scope, including the development of my bachelor's and master's theses.

Responsibilities included dataset preparation and cleaning, model training and evaluation, and participation in ML paper reading groups. Focused on computer vision for sports video understanding: player detection, segmentation, re-identification.

Artificial Intelligence Course for High School Students | JCMM

Sep 2024 - Jan 2026 (Brno, Czech Republic), Contract

Led two out of eight sessions of the [Artificial Intelligence: Principles, Prediction, Security course](#) focused on hands-on ML engineering, in which students used Jupyter notebooks to explore neural networks, including training a model for digit recognition. The course ran in the 2024/2025 and 2025/2026 academic years.

Course Assistant | Masaryk University

Sep 2021 - Dec 2021 (Remote), Contract

Reviewed and graded students' Python homework assignments in an introductory course IB111.

Education

Artificial intelligence and data processing, Master's degree | Masaryk University

2023 - 2026 (finished)

Master's thesis: **Camera view transition**: A design and implementation of a method for creating transitions between two camera views in soccer broadcasts. [Project website](#).

Informatics, Bachelor's degree | Masaryk University

2020 - 2023

Bachelor's thesis: **Event Synchronization in Multiple Video Recordings**: Implementation of VideoSnapping method for video synchronization with additional support for various descriptors to detect image features and evaluation on soccer videos. [Code](#), [paper](#).

Projects

My completed projects of various scopes can be found on my [website](#). Selected projects:

SAFE: Sandboxed Actions Forensics and Evaluation for AI Control via Action-Aware Monitoring

Mar 2026

Project done during [Apart's AI Control Hackathon](#). A proof-of-concept demonstration that monitoring of untrusted models was improved when the monitor could run the proposed commands in a sandbox and inspect the activity logs, such as file changes or network logs; [link](#).

Great Filter

Jul 2025 - Nov 2025

A Chrome extension for AI-powered content filtering on social media, developed in part to try out Claude Code; [link](#). Around 30 active users.

Blog @ Substack

2022 - present

Mostly AI-related [Substack](#), where I wrote about [evaluating LLMs to estimate the age of acquiring Czech words](#), explained [AI safety papers](#), or created a [reinforcement learning reference](#).

Volunteering

Blue Dot's AI Safety Fundamentals Facilitator

2023 - 2024 (Brno, Czech Republic)

Part of a two-person team facilitating three iterations of BlueDot AI safety courses in Brno.